

SRAM Scaling Opportunities Below $0.01\ \mu\text{m}^2$ Using Double-Row CFET Architecture with Wordline-Folded Bitcell Design for Performance Optimization

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Abstract

This work presents a DTCO exploration that enables SRAM scaling below $0.01\ \mu\text{m}^2$ in CFET nodes, achieving over 50% scaling compared to the A14 nanosheet node. It focuses on simplifying SRAM bitcell internal routing in CFET architecture, identifying process integration modules required for a compact CFET bitcell, and proposing design approaches to optimize performance. The DTCO study demonstrates that designing two bitcells together in a wordline-folded configuration, compared to bitcells without WL-folding, improves performance by $2\times$, enhances write margin by 162 mV, and enables a standard cell-like layout.

Introduction

SRAM bitcell area scaling faces challenges during the transition from the FinFET era to the first nanosheet device node, primarily because the sheet width exceeds the fin width, assuming other design parameters remain unchanged (Fig. 1). In nanosheet (NS) device nodes, architectural innovations such as forksheet (FS) devices can enable further scaling by replacing n-n and p-p or n-p separations with dielectric walls and eliminating gate extensions [1]. The CFET architecture [2][3][4] provides a promising path for SRAM scaling by stacking nFET over pFET devices. However, achieving a compact footprint for SRAM, given its intricate internal routing requirements, necessitates additional process steps.

This work presents an SRAM bitcell design in a double-row CFET (DR CFET) architecture [4], which enables effective cell height reduction and eliminates the traditional cross-coupling approach in CFET SRAM, thereby reducing process complexity. Our Design-Technology Co-Optimization (DTCO) study shows that designing two SRAM bitcells together in a wordline (WL)-folded configuration, improves performance and write margins while achieving a standard-cell-like bitcell with the same height and matching power and signal tracks. Further DTCO exploration demonstrates that SRAM bitcells in A7 CFET node can achieve an area as small as $0.009\ \mu\text{m}^2$.

Double-row CFET SRAM DTCO

The DR CFET architecture (Fig. 2) [4] is a promising choice for advanced nodes beyond nanosheet node, offering effective cell height reduction and reduced Middle-of-Line (MOL) process complexity [4]. The DR CFET SRAM (Fig. 3 (a), 4 (a)) comprises two rows, each forming a complete bitcell. Cross-coupling is implemented in the M0 metal layer (Fig. 3, 4 (a), top-tier views) without additional process steps, greatly simplifying integration vs. other implementations [5]. With M0 tracks allocated to cross-coupling and wordline picking, bitlines are routed in the backside metal (BM0). The bottom pFET device is used as the pass-gate transistor near the bitline in BM0, which removes the need for tall vias that would otherwise be required if the top nFET device were used instead. Top and bottom devices are connected through vertical drain-to-drain vias integrated within replacement metal (VMM) (Fig. 3(c)). Fig. 5 shows a schematic process flow to realize the top-bottom connection with inbetween the S/D region simulated with the SEMulator3D [6].

In our approach, the top nanosheet of the unused nFET device, which is shared with the neighboring cell, is selectively removed using a top-nanosheet-cut module (Fig. 3(b)). Unlike single and double diffusion breaks (Fig. 6), this approach enables shared S/D contacts for bitline access between adjacent cells while preserving the gate structure, thereby allowing front-side WL access from the bottom pFET pass-gate. The resulting bitcell achieves a height of 69 nm and a width of $4\times\text{CPP}$, yielding an area of $0.011\ \mu\text{m}^2$ (Fig. 1).

Design Co-optimization: The design shown in Fig. 3(a) requires six backside metal tracks for the pair of bitlines (BL and BLB) corresponding to the two bitcells, and VDD. This configuration results in narrower bitline metal for the given cell height, increasing bitline resistance, which degrades both performance and write margin. Additionally, routing the bitlines in proximity without shielding metal between them makes the bitcell more susceptible to noise due to bitline coupling. To address these challenges, a WL-folded bitcell design is proposed, as shown in Fig. 4(a), where two bitcells are designed together by sharing the bitlines and different WLs. This design improves performance and write margin by reducing bitline resistance (achieved by increasing BL width) and introducing shielding metal between the bitlines that minimizes bitline coupling. This modification also aligns the bitcell with the DR CFET logic standard-cell rules, matching the height and power/signal tracks (Fig. 2) [4].

The bitcell height can be further scaled by removing the VSS power-wall and routing the VSS on M0 metal and maintaining a wider VDD at BM0 (Fig. 4(b)). Since the pass-gates are pFET devices, VDD resistance is more critical than VSS for overall performance. Eliminating the VSS power-wall enables gate-cut scaling to 12 nm, compared to the wider gate-cut width required for VSS power-wall placement. These modifications result in a 21.7% scaling in cell height compared to the design in Fig. 4(a) and a 53% area reduction from the A14 NS bitcell (Fig. 1).

DTCO Flow and Results: Design rules from Table 1 were applied to create a 3D structure (Fig. 7) with technology-specific inputs using SDE [7] for each SRAM bitcell option. RC parasitic extraction and netlist generation were performed using Raphael FX [7], while the channel region was modeled with a TCAD-calibrated BSIM-CMG model [4]. The extracted bitcell netlist was then utilized for stability and array-level performance evaluations using Spectre [8]. Monte Carlo (MC) simulations analyzing nanosheet width variation indicate that the mean hold and read static noise margins (HSNM and RSNM) are comparable across designs. The proposed bitcell design, which places all devices within the same nanosheet row and vertically stacks nFET and pFET devices, results in correlated nanosheet width variations due to shared process conditions. Simulation results incorporating these correlations demonstrate reduced 3σ HSNM and RSNM values compared to uncorrelated cases (Fig. 8).

The WL-folded design approach improves read delay and write margin by $2\times$ and 162 mV, respectively (Fig. 9), due to reduced bitline resistance (Fig. 10) compared to the bitcell without WL folding. Additionally, the scaled WL-folded SRAM offers enhanced performance by reducing WL capacitance (Fig. 10). However, the scaled design exhibits a slight degradation in write margin (Fig. 9), attributed to increased bitline resistance in the narrower BM0 metal. The standard-cell-like SRAM bitcell eliminates the need for a keep-out zone (KOZ) between the array and periphery/logic, leading to a lower effective area per bit in smaller SRAM macros compared to those using scaled SRAM bitcells, which require KOZ (Fig. 11).

Conclusions

This work presented an SRAM bitcell design in a DR CFET architecture that enabled SRAM scaling to $0.009\ \mu\text{m}^2$ at A7 CFET node with greatly simplified internal routing. The DTCO study highlighted that designing two bitcells together using a wordline-folded approach improved read delay by $2\times$ and write margin by 162 mV, while also enabling a standard cell-like bitcell layout for better logic integration.

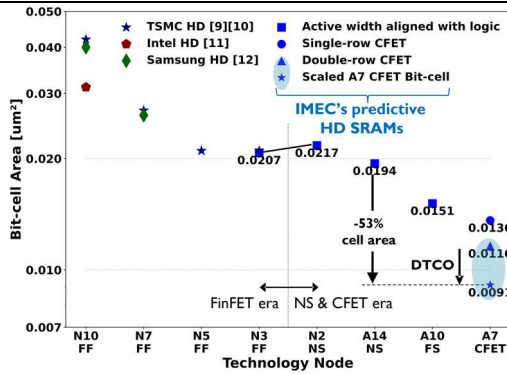


Fig. 1: High-Density SRAM bitcell area scaling projections crossing from the commercial FinFET era to the predictive nanosheet and CFET era.

TABLE I: IMEC'S TECHNOLOGY NODES AND DESIGN PARAMETERS

Parameters	N2 NS	A14 NS	A10 Forksheet	A7 CFET
CPP (nm)	48	45	42	42
Gate length (nm)	15	15	14	14
Active width (nm)	18	15	17	19
MMP (nm)	22	21	18	19/18

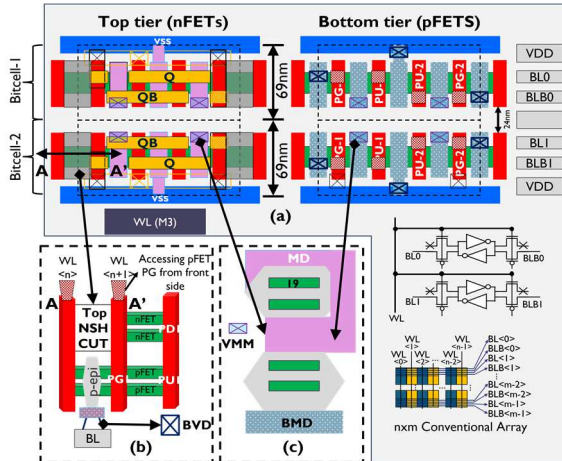


Fig. 3: (a) Top and bottom tier layout of DR CFET SRAM **without WL-folding**. Bitcells are arranged in rows with BL and BLB (see inset). Six BM0 tracks are needed for bitlines and VDD, causing narrow bitlines and high BL-BLB coupling due to proximity. (b) Cross-sectional view along A-A', showing selective removal of the unused nFET nanosheet, retaining the gate for WL access from the bottom pFET pass-gate. (c) Cross-sectional view depicting direct drain-to-drain connection (VMM) forming inverter outputs.

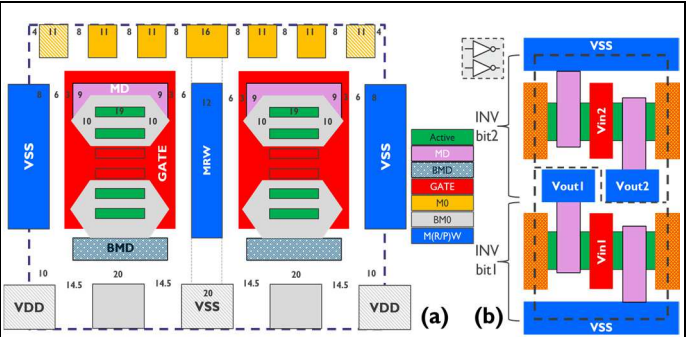


Fig. 2: (a) Cross-sectional view of DR CFET cell architecture [4], which consists of two pairs of CFETs (four FETs in total) with a shared single middle-row wire (MRW) in the middle outer zone, achieving a total cell height of 7T. It features 7 M0 and 2 BM0 signal routing tracks on the frontside and backside, respectively. The two outer M0 tracks are utilized for PnR, while three BM0 tracks are reserved for global VDD and VSS rails. Additionally, the DR CFET incorporates a local VSS wall at the cell edges, shared with adjacent cells, with PD and PU networks routed through frontside and backside metal-via layers, respectively. Vertical MRWs provide the connection between both networks. (b) Layout view: an example of cell-level optimizations within the DR CFET architecture, demonstrating the creation of a 2-bit INV21 cell [4].

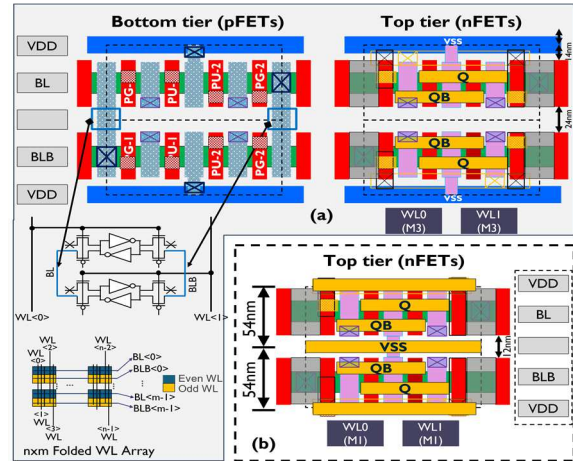


Fig. 4: (a) Bottom and top tier layout of DR CFET SRAM **with WL-folding**. Two bitcells share bitlines with separate wordlines (WL0, WL1), reducing bitline resistance, enhancing performance, and minimizing BL-BLB coupling through shielding. This aligns the bitcell with DR CFET logic rules (Fig. 2(b)). (b) Scaled DR CFET SRAM bitcell with WL-folding, achieved by removing the VSS power wall and routing VSS via M0, enabling middle gate-cut scaling to 12 nm.

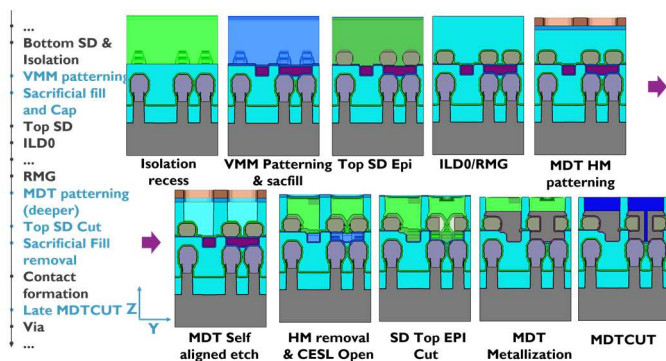


Fig. 5: Process flow to create top-bottom connection with minimal extension (individual connection per device). This inbetween connection is formed by pre-patterning between the formation of the bottom source-drain and the top source-drain. The contact patterning is etched deep enough to reach the sacrificial connection pattern which is then removed by isotropic etch. To reduce the needed extension and increase the connectivity the lateral size of the EPI region is trimmed with a mask, the contact is formed, and the contact trench is filled. The last step is a late metal cut; moving the cut later in the flow reduces the amount of liner in the top-down connection.

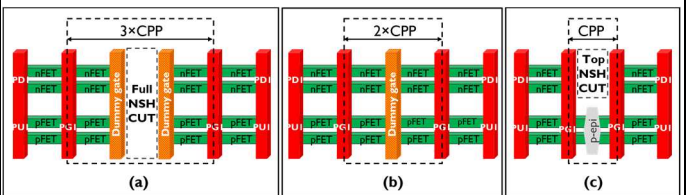


Fig. 6: Unused device nanosheet stop options showing half-cells of two adjacent bitcells in CFET architecture: (a) **Double diffusion break**, where both top and bottom nanosheets are cut using a nanosheet cut mask, preventing bitline S/D contact sharing. This requires a 3× CPP distance between adjacent pass-gates. (b) **Single diffusion break**, where the top and bottom nanosheets are cut and replaced with a dummy gate. In this case as well, bitline S/D contact cannot be shared, requiring a 2× CPP distance between adjacent pass-gates. (c) **Top-nanosheet-cut**, where only the top nanosheets of the unused nFET device are selectively removed while preserving the monolithic gate, allowing WL pickup from the bottom pFET pass-gates and enabling bitline S/D contact sharing between adjacent bitcells with a minimal CPP distance.

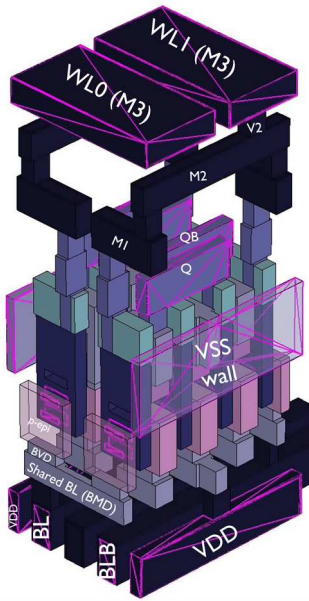


Fig. 7: 3D structure of the DR CFET SRAM bitcells with a WL-folded design, where two bitcells are designed together, sharing the bitlines (BL and BLB on backside metal BM0) and having separate wordlines (WL0 and WL1 on M3). The structure is built using A7 technology inputs, including materials and dimensions, through SDE [7], with bitcell parasitic extraction conducted using Raphael-FX [7].

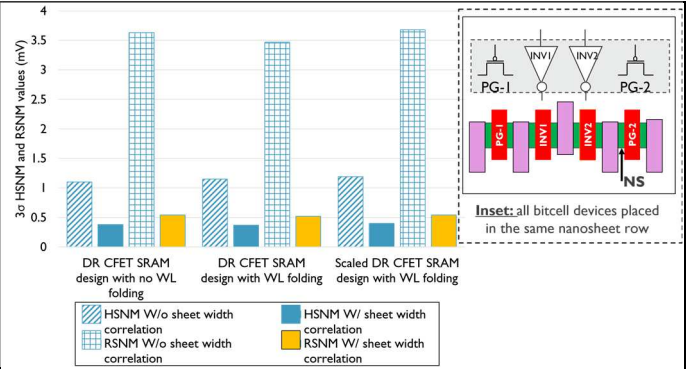


Fig.8: Comparison of 3σ hold and read static noise margins under nanosheet width variations from Monte Carlo simulations with and without correlation among the six bitcell devices. The results show lower 3σ margins when correlation is considered, highlighting the impact of device proximity and stacking in the proposed bitcell. **The inset** provides a bitcell layout view, demonstrating that all six transistors are positioned within the same nanosheet row and stacked to form inverters in the CFET architecture, supporting the assumption of correlated nanosheet width variations. The mean HSNM and RSNM values are similar across designs (~ 310 mV and ~ 145 mV, respectively).

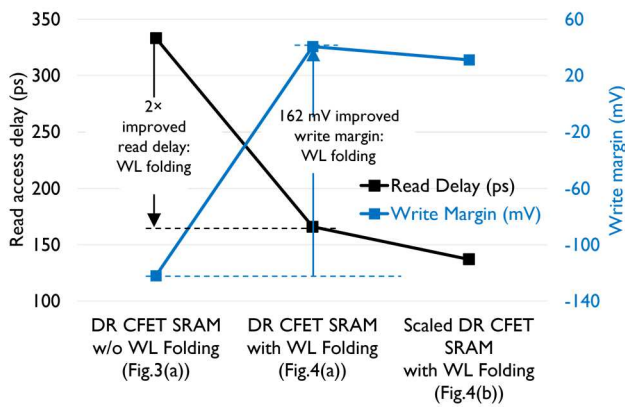


Fig.9: Simulation results for read access delay and write margin for the three DR CFET SRAM designs. The WL-folding approach achieves a 2 $\times$ improvement in read delay and enhances the write margin by 162 mV.

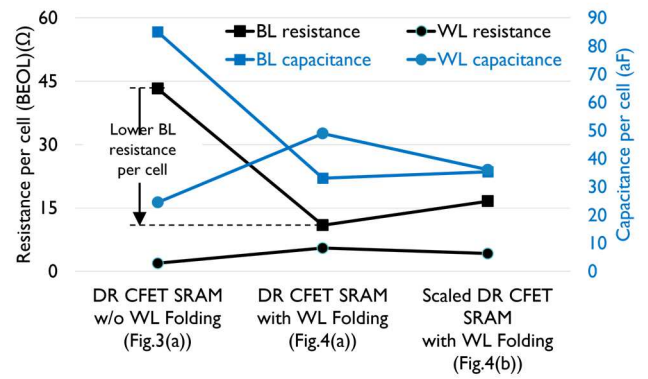


Fig.10: BEOL resistance and total capacitance per cell for the three DR CFET SRAM bitcell designs. The WL-folding design effectively reduces bitline resistance per cell, contributing to improved performance and write margin.

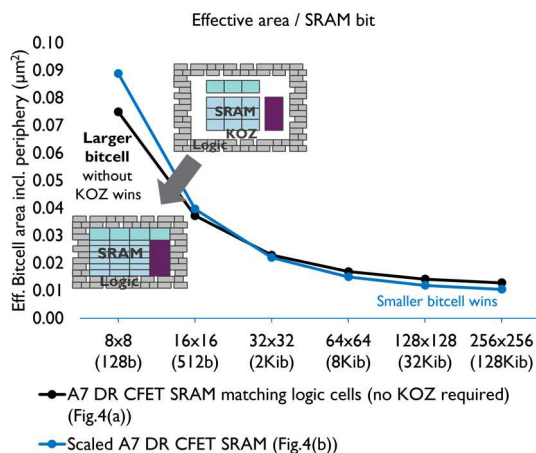


Fig. 11: Comparison of effective bitcell area per bit (total SRAM macro area / macro size). The standard-cell-like SRAM bitcell eliminates the need for a KOZ between the bitcell array and periphery/logic, resulting in a lower effective area per bit for smaller SRAM macros. In contrast, SRAM macros with scaled bitcells require KOZ, leading to higher area overhead at smaller macro sizes. As the macro size increases, the SRAM macros with scaled bitcells eventually become more area efficient.

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